PureData

PureData, or PD, is quite the departure from the text-based programming of Arduino and Processing. PD is a graphical programming environment where lines of code and functions are replaced by reusable objects that are connected together with actual lines that link one object to another—kind of like the patches from an analog synthesizer. PD is particularly adept at audio, image, and video manipulation, and is a loosely related, free and open-source alternative to Max/MSP. If you are interested in using the Arduino to interface with sound and video generated on a computer, then PD would be a good choice. More information and downloads can be found at `http://puredata.info`.

Figure 11-14 shows a simple PD patch that reads sensor data from an analog sensor connected to pin A0 on an Arduino board with the SimpleAnalog version of Firmata loaded on it. It then uses this sensor data to control the amplitude and frequency of a band pass filter applied to a static noise signal.

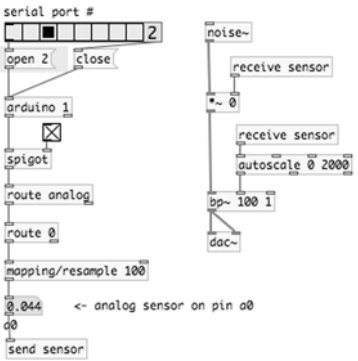


Figure 11-14. *Simple PureData patch*

This simple patch demonstrates some of the basic capabilities of PD and gives you a taste for what it's like to program in a graphical language. For this patch to work, we need to download the Pduino object available from `http://at.or.at/hans/pd/objects.html`. For some help getting started with Pduino, check out the FLOSS Manual on PureData at `http://en.flossmanuals.net/pure-data/ch061_starting-pduino`. Included with the Pduino download are several example files, including `arduino-test.pd`, shown in Figure 11-15.